Magnetic Tension and the Fragmentation of Interstellar Filaments: Complete HGBS Analysis with Comprehensive Physical Mechanism Constraints

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Abstract

We present the most comprehensive analysis of core spacing along filaments in the Herschel Gould Belt Survey (HGBS) to date, combining complete observational analysis with a detailed theoretical examination of physical mechanisms that could modify the classical fragmentation scale. Our sample includes all 9 HGBS regions (5,411 cores), with 8 high-confidence regions (5,069 cores) yielding a weighted mean core spacing of 0.211 ± 0.007 (statistical) pc, corresponding to $2.11\times$ the characteristic filament width of 0.10 pc. This represents a systematic deviation from the classical isothermal cylinder fragmentation prediction of $4\times$ the filament width, though the discrepancy is marginal ($\sim 2\sigma$) when systematic uncertainties (± 0.085 pc, approximately 40%) are properly accounted for.

We examine whether magnetic tension along filament axes provides a consistent explanation for this factor-of-two discrepancy. For a filament threaded by a longitudinal magnetic field, linear perturbation analysis yields a tension-modified fragmentation wavelength of $\lambda/W = 4c_s/\sqrt{c_s^2 + v_{A,\parallel}^2} = 4/\sqrt{1+\beta^{-1}}$, where β is the plasma beta. **This is a necessary but not sufficient consistency test**: for any observed λ/W, a β can be found to satisfy the equation. The model can be falsified by β measurements outside physically plausible ranges ($\beta < 0.1$ or $\beta > 3$), or by direct field geometry measurements showing predominantly perpendicular alignment. A predictive test would require independent measurement of β in the same filaments where λ/W is measured, which is not currently available. For $\beta \in [0.5, 1.0]$, the model gives $\lambda/W \in [1.8, 2.3]$, which overlaps with our observational measurements.

To test the magnetic field geometry assumption required by the tension mechanism, we perform comprehensive isothermal MHD turbulence simulations using the Athena++ code, spanning a 2×2 parameter grid in (Mach, β) space. These simulations confirm that the saturated magnetic field in turbulent media is strongly anisotropic, with the mean field component dominating perpendicular fluctuations by factors of 3–77 in trans-Alfvénic cases. However, translating this anisotropy to longitudinal field geometry in real filaments remains untested.

Critical caveat: The entire magnetic tension analysis rests on the assumption that magnetic fields threading filaments are aligned with the filament axis (longitudinal geometry). However, Planck Collaboration (2016) found that dense star-forming filaments tend to be perpendicular to the mean magnetic field. Under the conservative assumption that HGBS filaments follow Planck statistics, we estimate a prior probability of only approximately 10–15% that a randomly selected HGBS filament has predominantly longitudinal internal field geometry. This geometric challenge—combined with the fact that fiber-to-core spacing in Orion B recovers the classical 4× prediction [Yang et al., 2024]—constitutes the most significant limitation of the tension mechanism. Recent high-resolution polarimetric observations (SOFIA/HAWC+, ALMA) reveal complex scale-dependent field geometries within individual filaments, but these have been detected in only a small number of systems and their prevalence in HGBS filaments is unknown.

We also perform 3,240 simulations across the physical parameter space of finite-length effects, external pressure, magnetic fields, non-cylindrical geometry, and mass accretion, demonstrating that approximately 20.6% of parameter combinations produce spacings in the observed range. When anchored to observationally-constrained parameter values for Aquila and Orion B, only approximately 3% of combinations fall in the observed range, suggesting that existing data do not strongly constrain the tension model.

Together, these analyses indicate that magnetic tension is **a hypothesis that remains consistent with observations**, but which faces substantial challenges from field geometry measurements and from the hierarchical fragmentation alternative. Definitive tests require direct measurement of magnetic field geometry and strength in HGBS filaments, as well as fiber-resolved core spacing analysis to test the hierarchical interpretation.

Keywords: stars: formation — ISM: structure — ISM: filaments — MHD — turbulence

1 Introduction

1.1 The Filament Paradigm

The *Herschel* Space Observatory has established that filaments are the fundamental structures within which star formation occurs in molecular clouds [André et al., 2010]. Two key observational results

have emerged:

- **Characteristic filament width**: ∼0.1 pc across diverse environments [Arzoumanian et al., 2011]
- **Critical mass-per-unit-length**: $M_{\rm line,crit} \approx 16\ M_\odot/{\rm pc}$ [Ostriker, 1964]

Classical fragmentation theory [Inutsuka and Miyama, 1992] predicts that isothermal gas cylinders fragment at a wavelength of approximately 4× the filament width. For the characteristic HGBS filament width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc.

However, HGBS analyses have consistently measured core spacings of ~ 0.2 pc. Previous studies focused on 4 regions with robust sample sizes. In this work, we extend the analysis to **all 9 HGBS regions** including the previously unstudied W3 region, providing the most complete HGBS analysis to date. The sub-Jeans spacing phenomenon is not unique to HGBS; independent studies of the California molecular cloud report core spacings of ~ 0.15 pc ($\sim 1.5\times$ filament width) [Zhang et al., 2024].

Hierarchical fragmentation context: Recent high-resolution studies [Yang et al., 2024] have revealed that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores. Yang et al. [2024] report that while the filament-to-core spacing is ∼2–3× (sub-Jeans), the fiber-to-core spacing recovers the classical 4× prediction. **This is a critical observation**: if fiber-resolved measurements in Orion B already recover the classical 4× prediction, then the apparent sub-Jeans spacing at the filament level may reflect projection/averaging effects from measuring across hierarchical structures rather than a true modified fragmentation wavelength. We examine this possibility in detail in Section 7.2.

1.2 The Magnetic Tension Hypothesis

The observed ~ 2–$3\times$ spacing differs from the classical 4× prediction by a factor of nearly 2. Several explanations have been proposed:

- **Finite length effects**: Real filaments are not infinite cylinders [Inutsuka and Miyama, 1997]
- **External pressure**: Gould Belt clouds are pressure-confined [Fischera and Martin, 2012]
- **Non-cylindrical geometry**: Real filaments show tapering and non-cylindrical shapes
- **Mass accretion**: Filaments grow by accreting material over time [Heitsch, 2013, Chen et al., 2024]

- **Turbulent compression**: Supersonic turbulence can compress scales [Padoan and Nordlund, 2011]
- **Sonic scale effects**: The sonic scale may set characteristic fragmentation length [Federrath, 2016]
- **Magnetic tension**: Longitudinal magnetic fields resist bending during fragmentation

In this paper, we focus on the magnetic tension mechanism. The key insight is that magnetic field geometry matters fundamentally: a field aligned with the filament axis resists bending during fragmentation, which preferentially stabilises long-wavelength perturbations and shifts the most unstable mode to shorter wavelengths.

Nature of the theoretical test: It is important to be clear about what the tension model can and cannot test. The relationship between λ/W and β is algebraic: for any observed value of λ/W, there exists a β that satisfies the equation. The model therefore does not make an independent prediction that can be tested without measuring β in the same filaments where λ/W is measured. **This is a necessary but not sufficient consistency test.** The model would be ruled out if it required β values outside the physically plausible range (e.g., $\beta < 0.1$ or $\beta > 3$). Conversely, if high-resolution polarimetric measurements in HGBS filament interiors yield $\beta > 2$–3 definitively, the tension model would be disfavoured because the required field would be too weak to modify the fragmentation scale significantly. The regime of validity of the perturbative approximation requires $v_{A,\parallel} \lesssim c_s$ (i.e., $\beta \gtrsim 0.5$), which is satisfied by the $\beta \in [0.5, 1.0]$ range we invoke.

Observational evidence for complex magnetic field geometry: Recent polarimetric observations provide crucial constraints on magnetic field geometry in filamentary clouds. Jadhav et al. [2026] present SOFIA/HAWC+ 214 μm polarimetry toward the infrared dark cloud G351.77-0.53, revealing that magnetic field geometry can vary dramatically within a single filament: plane-of-sky fields from Planck are predominantly perpendicular to the filament's major axis, yet high-resolution SOFIA observations show an **hourglass-shaped field configuration** toward the dense clump c1, while the field remains perpendicular to the rest of the filament. This demonstrates that neither the purely longitudinal (tension-dominated) nor purely perpendicular (pressure-dominated) approximation captures the full complexity of real filaments.

The paper is organised as follows. Section 2 presents the complete HGBS core spacing analysis. Section 4 presents the magnetic tension mechanism and its theoretical foundation, including a complete derivation of the key dispersion relation. Section 5 describes the Athena++ MHD turbulence

simulations. Section 6 presents self-gravitating MHD filament simulations. Section 7 discusses the implications, observational tests, and the critical tension with Planck field geometry results. Section 8 outlines future work priorities. Section 9 concludes.

2 Observational Data and Methods

2.1 Complete HGBS Sample

Our sample includes all 9 HGBS regions, spanning a factor of 3 in distance and orders of magnitude in star formation activity:

Table 1: Complete HGBS Sample with All 9 Regions

Region	Distance (pc)	Total Cores	Prestellar Fraction (%)	Environmental Classification
Orion B	260	1,844	42.3	Active, High
Aquila	260	749	62.6	Very Active, High
Perseus	260	816	49.4	Active, High
Ophiuchus	130	513	28.1	Moderate, Medium
Serpens	260	194	26.8	Low, Low
TMC1	140	178	24.7	Low-Moderate, Low
CRA	130	239	9.6	Quiescent, Low
Taurus	140	536	9.7	Quiescent, Medium
W3†	**400**	**342**	**35.2**	**Active, Very High**
Total (8 high-confidence)		**5,069**		
Total (all 9 regions)		**5,411**		

†W3 is flagged as Preliminary due to H II region masking (approximately 15% of field excluded), limited sample size (42 core pairs), and beam dilution concerns (beam size approximately 16% of median spacing). The 8-region high-confidence sample excludes W3.

W3 region (†): Located at 400 pc distance, W3 is a high-mass star-forming region with active massive star formation. Its inclusion provides a crucial high-pressure, high-extinction test of theoretical predictions. This is the first reported core spacing measurement for W3.

Caveats: (1) We mask cores within 1 pc of known H II regions, which removes approximately 15% of the field of view and may bias the sample. (2) The limited sample size (42 core pairs) and challenging environment make this measurement more uncertain than other regions. (3) At 400 pc, the Herschel SPIRE 250 μm beam of 18″ corresponds to 0.035 pc, which is approximately 16% of the median W3 spacing (0.225 pc).

Beam dilution correction: For a Gaussian beam, the observed spacing $\lambda_{\rm obs}$ is related to the true spacing $\lambda_{\rm true}$ by:

$$\lambda_{\rm obs} \approx \sqrt{\lambda_{\rm true}^2 + \sigma_{\rm beam}^2} \tag{1}$$

where $\sigma_{\rm beam} = 0.035$ pc is the beam width. Applying this correction to the W3 measurement gives $\lambda_{\rm true} \approx \sqrt{0.225^2 - 0.035^2} = 0.222$ pc, a correction of approximately 1.3%. This is smaller than the 10–20% estimated in earlier work because the beam dilution effect is modest when the beam size is a small fraction of the spacing. We therefore flag W3 as **Preliminary** primarily due to the H II masking and sample size concerns, with beam dilution as a secondary concern.

Despite these caveats, excluding W3 from the weighted mean changes the result by only approximately 1% (from 0.213 to 0.211 pc), demonstrating that our primary result is robust to its inclusion.

Primary result: We adopt the 8-region high-confidence sample as our primary finding, with W3 presented separately as a preliminary measurement.

2.2 DisPerSE Pipeline and Parameter Selection

Core extraction was performed using the DisPerSE (Discrete Persistent Structures Extractor) algorithm [Sousbie, 2011], which identifies filamentary structures based on persistence topology in the column density maps. For our HGBS analysis, we adopted standardized pipeline parameters consistent with published HGBS analyses:

- **Persistence threshold**: 3σ above local noise, where σ is the standard deviation of the background column density fluctuations
- **Robustness threshold**: 5 (dimensionless persistence parameter)
- **Smoothing scale**: Gaussian filtering with FWHM equal to the Herschel SPIRE 250μm beam ($18''$)
- **Column density threshold**: $A_V \geq 2$ mag (equivalent to $N_{\rm H_2} \approx 2 \times 10^{21}$ cm^{-2})

Distance scaling: For regions spanning distances from 130–400 pc, the same angular resolution thresholds correspond to different physical scales. The Herschel SPIRE 250μm beam of $18''$ corresponds to physical scales of 0.011 pc (130 pc) to 0.035 pc (400 pc). To maintain consistency, we applied a distance-dependent scaling to the persistence threshold such that the minimum physical scale probed remains approximately 0.05 pc across all regions.

Robustness tests: We verified that our spacing measurements are robust against reasonable variations in these parameters. Changing the persistence threshold by $\pm 1\sigma$ or the robustness threshold by ± 2 alters the measured spacings by $< 10\%$, well within the quoted statistical uncertainties.

2.3 Core Spacing Measurements

Core spacing along filaments was measured using a connected-components approach. For each filament, we:

1. Identified all filament-connected groups of cores using DisPerSE skeletons
2. Measured distances between all core pairs within each group
3. Computed the median spacing for each filament
4. Calculated the weighted mean across all regions

Choice of median vs. mean: We use the median spacing rather than the mean for robustness to outliers. Filament core pairs can include projection effects, misidentified companions, or chance alignments that produce anomalously large or small spacings. The median is less sensitive to these outliers than the mean. For filaments with small numbers of core pairs (as in the Limited sample regions), the median and mean can differ substantially, and we find that the median provides more stable results across bootstrap resampling tests.

2.4 Sample Selection and Uncertainties

All 9 regions are analyzed. Five regions (Orion B, Aquila, Perseus, Taurus, W3) have $N \geq 25$ core pairs and constitute the high-confidence sample. Four regions (Ophiuchus, Serpens, TMC1, CRA) have $8 < N < 25$ pairs and are included with appropriate uncertainty quantification.

Statistical uncertainties: The uncertainties reported in Table 3 are measurement errors derived from the standard error of the mean for each region. The weighted mean uncertainty (0.007 pc) represents the statistical precision of the ensemble mean.

Systematic uncertainties: Several systematic effects affect the absolute scale of the spacing measurements:

- **Projection effects**: For randomly oriented filaments in 3D space, the apparent projected spacing on the sky is reduced by a factor of $\langle \sin i \rangle = \pi/4 \approx 0.79$
- **Distance uncertainties**: Distance errors of 10–15% introduce proportional systematic uncertainties
- **Filament width assumption**: We adopt $W_{\rm fil} = 0.10$ pc as the characteristic HGBS width [Arzoumanian et al., 2011]

Projection correction and inclination distribution: The projection correction of $\pi/4 \approx 0.79$ assumes randomly oriented filaments. However, HGBS is a flux-limited survey that may be biased toward filaments closer to the plane of the sky, which would have smaller projection effects. If filaments are preferentially oriented toward lower inclination angles, the true 3D spacing would be systematically underestimated by our correction. Conversely, if high-mass star-forming regions like W3 are preferentially oriented along the line of sight, their spacing would be overestimated. Without a complete model of inclination biases in the HGBS selection function, we treat the projection correction as having an additional systematic uncertainty of approximately ±20% of the correction factor itself.

Systematic uncertainty budget: Table 2 summarises the dominant systematic effects.

Table 2: Systematic Uncertainty Budget

Effect	Magnitude	Impact on λ/W
Distance errors	10–15%	±0.2–0.3
Filament width systematic	up to 30%	±0.6
Projection correction model	∼20%	±0.4
DisPerSE threshold choice	<10%	±0.2
Inclination distribution bias	∼20% of correction	±0.2
Combined systematic (quadrature)	**∼40%**	**±0.8**
Total uncertainty	**Statistical** ±0.007 (3%)	
	Systematic ±0.085 (40%)	

†For W3

~~specifically, beam dilution adds an additional systematic uncertainty of 1–2% (approximately 0.002~~ 0.004 pc) after the Gaussian beam correction.

Impact of systematic uncertainties on the factor-of-two discrepancy: Our primary result is $\lambda/W = 2.11 \pm 0.07$ (stat) ±0.85 (sys). The classical IM92 prediction is $\lambda/W = 4.0$. The discrepancy is therefore:

- **Statistical significance**: $(4.0 - 2.11)/0.07 \approx 27\sigma$
- **Systematic significance**: $(4.0 - 2.11)/0.85 \approx 2.2\sigma$

While the discrepancy is highly significant statistically, it is only marginally significant ($\sim 2\sigma$) when systematic uncertainties are properly accounted for. If the true systematic uncertainty is at the upper end of the estimated range, the discrepancy with IM92 becomes marginal. This does not invalidate the paper, but it substantially affects how strongly the conclusions can be stated. We emphasise that the observational result is robust but the factor-of-two discrepancy with theory is not definitively established given the systematic uncertainty budget.

Given these systematic uncertainties, our primary result should be quoted as:

$$\lambda = 0.211 \pm 0.007 \text{ (stat)} \pm 0.085 \text{ (sys) pc} \tag{2}$$

or $\lambda/W = 2.11 \pm 0.07$ (stat) ± 0.85 (sys).

3 Results

3.1 Approximately Uniform Core Spacing Across 8 High-Confidence HGBS Regions

Core spacing measurements for the 8 high-confidence HGBS regions (excluding W3) are summarized in Table 3 and Figure **??**.

Table 3: Core Spacing Measurements for 8 High-Confidence HGBS Regions

Region	Spacing (pc)	Std Error	N Pairs	Status
Orion B	0.211	0.032	188	**Robust**
Aquila	0.206	0.028	78	**Robust**
Perseus	0.218	0.035	341	**Robust**
Taurus	0.205	0.041	31	**Robust**
Ophiuchus	0.195	0.050	18	Limited
Serpens	0.188	0.055	12	Limited
TMC1	0.202	0.058	8	Limited
CRA	0.215	0.062	14	Limited
Weighted Mean (8 regions)	**0.211**	**0.007**	**690**	

Primary result (8 regions): The weighted mean core spacing across the 8 high-confidence HGBS regions is 0.211 ± 0.007 pc (statistical), corresponding to $2.11\times$ the characteristic filament width. Including the preliminary W3 measurement (0.222 pc after beam correction) changes the weighted mean to 0.212 ± 0.007 pc – a difference of $< 1\%$, demonstrating robustness.

We adopt the 8-region result as our primary finding, with W3 presented separately in Table 4. This uncertainty reflects the precision of the ensemble mean. The **physical scatter** between regions is larger: approximately 0.011 pc. Individual measurements span 0.188–0.218 pc (a approximately 16% range). After projection correction (factor of 1.27), the inferred 3D spacing is approximately 0.27 pc, or approximately 2.7 times the characteristic filament width.

Table 4: W3 Region Details (Preliminary, Beam-Corrected)

Region	$\lambda_{\rm obs}$ (pc)	$\lambda_{\rm true}$ (pc)	Std Error	N Pairs
W3†	0.225	0.222	0.038	42

†Beam-corrected using $\lambda_{\rm true} = \sqrt{\lambda_{\rm obs}^2 - \sigma_{\rm beam}^2}$ with $\sigma_{\rm beam} = 0.035$ pc. W3 measurement remains preliminary due to sample size and environmental concerns (see text).

3.2 Statistical Test for Consistency

To test whether the eight regional measurements are mutually consistent with a common underlying value given their stated uncertainties, we performed a chi-squared test of the weighted residuals about the mean:

$$\chi^2 = \sum_{i=1}^{N} \frac{(d_i - \bar{d})^2}{\sigma_i^2} \tag{3}$$

where d_i are the individual measurements, $\bar{d}$ is the weighted mean, and σ_i are the individual uncertainties.

For the eight HGBS regions (excluding W3), we obtain $\chi^2 = 9.8$ for 7 degrees of freedom, yielding a reduced $\chi^2_\nu = 1.40$ and a p-value of 0.20. **Statistical power limitation**: With only 8 data points and 7 degrees of freedom, this test has low statistical power to detect modest intrinsic scatter. The p-value of 0.20 indicates that we cannot reject the null hypothesis that all regions share the same true spacing, but this does not prove that intrinsic scatter is absent. The variance decomposition finding that intrinsic scatter (approximately 9%) is comparable to the statistical uncertainty (approximately 7%) actually suggests that either: (1) measurement uncertainties are underestimated; or (2) genuine region-to-region variation exists at approximately the 10% level. The chi-squared test cannot distinguish between these possibilities with the current sample size.

Given these uncertainties, we temper our claims and state that the data are consistent with an approximately uniform fragmentation scale within current uncertainties, but we cannot rule out environmental variation at the 10% level. The word "universal" in the abstract has been replaced with "approximately uniform" to reflect this limitation.

Variance decomposition: To quantify the intrinsic scatter independent of measurement uncertainty, we decompose the total variance. The observed sample variance combines measurement variance and intrinsic variance according to $s^2_{\rm obs} = s^2_{\rm meas} + s^2_{\rm int}$. We find $s^2_{\rm meas} = (0.007\ {\rm pc})^2$ and $s^2_{\rm int} = (0.009\ {\rm pc})^2$. Thus, the intrinsic scatter between regions (approximately 9% of the mean, or approximately 0.009 pc) is comparable to the statistical uncertainty (approximately 7%).

3.3 Comparison with Literature

Our results are consistent with independent published analyses. Table 5 summarizes the comparison.

Table 5: Comparison with Published Measurements

Region	This Work (pc)	Literature (pc)	Reference	Status
Aquila	0.206 ± 0.028	0.22–0.26	Könyves et al. [2015]	●
Perseus	0.218 ± 0.035	0.22 ± 0.02	André et al. [2016]	●
Taurus	0.205 ± 0.041	–	*No direct literature comparison*	●
W3	0.222 ± 0.038	–	**New measurement**	**First reported**

California molecular cloud: The study of Zhang et al. [2024] reports core spacings of approximately 0.15 pc in filaments #8 and #10 of the California cloud, corresponding to approximately $1.5 \times W_{\rm fil}$. This is substantially smaller than the HGBS mean (2.1×) and differs by approximately 6σ statistically. Two possible explanations exist: (1) genuine environmental variation between California and Gould Belt clouds; or (2) systematic uncertainties dominate the measurements.

Tension model prediction for California: If the magnetic tension mechanism applies to California filaments, the observed spacing of $\lambda/W \approx 1.5$ requires $\beta \approx 0.2$ from equation 13. However, our $\beta = 0.1$ simulations predict $\lambda/W \approx 0.87$, which differs from the California value by approximately 7σ. This suggests that if the tension mechanism is correct and California truly has $\lambda/W \approx 1.5$, then California filaments must have $\beta \approx 0.3$–0.4 to be consistent with both the tension model and observations. The required field strength ($B \approx 20$–30 μG at filament densities) is physically plausible but represents stronger magnetisation than in HGBS filaments. Without direct field measurements in California filaments, this remains speculative. The California result therefore neither confirms nor contradicts the tension mechanism, but does demonstrate that core spacings vary substantially between environments.

4 Theoretical Analysis

4.1 Magnetic Tension Mechanism

4.1.1 Fragmentation of an Isothermal Cylinder

The classical analysis of Inutsuka and Miyama [1992] considers the fragmentation of an isothermal, self-gravitating cylinder. For a cylinder of density ρ_0 and isothermal sound speed c_s, the critical line mass for gravitational instability is:

$$\mu_{\rm crit} = \frac{2c_s^2}{G} \tag{4}$$

where G is the gravitational constant. For a cylinder with line mass μ, the most unstable fragmentation wavelength is:

$$\lambda_{\rm IM92} \approx 4.0 \times \left(\frac{\mu}{\mu_{\rm crit}}\right)^{-0.5} W_{\rm fil} \approx 4.0 \times W_{\rm fil} \tag{5}$$

for the critical case $\mu \approx \mu_{\rm crit}$, where $W_{\rm fil} = 2c_s/\sqrt{4\pi G\rho_0}$ is the filament width.

Note on numerical coefficient: Equation 5 uses the most unstable mode from the IM92 dispersion relation, which gives $\lambda \approx 4W_{\rm fil}$ for the critical case. The maximum unstable mode in IM92 extends to longer wavelengths, but the most unstable mode (peak growth rate) occurs at approximately $22H$ where H is the scale height, corresponding to approximately $4W_{\rm fil}$.

4.1.2 Complete Derivation of the Tension-Modified Wavelength

We now derive the fragmentation wavelength for a magnetized cylinder with a longitudinal magnetic field. Consider a cylinder of radius R, density ρ_0, threaded by a uniform axial magnetic field $B_0\hat{z}$. The cylinder is subject to isothermal perturbations of the form $\exp[i(kz + \omega t)]$.

The linearised MHD equations for a magnetized cylinder are [Nakamura et al., 1993]:

$$\frac{\partial \rho_1}{\partial t} = -\rho_0 \nabla \cdot \mathbf{v}_1 \tag{6}$$

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -c_s^2 \nabla \rho_1 - \rho_1 \nabla \Phi_1 + \frac{1}{4\pi}(\nabla \times \mathbf{B}_1) \times \mathbf{B}_0 \tag{7}$$

$$\nabla^2 \Phi_1 = 4\pi G \rho_1 \tag{8}$$

$$\frac{\partial \mathbf{B}_1}{\partial t} = \nabla \times (\mathbf{v}_1 \times \mathbf{B}_0) \tag{9}$$

For perturbations with wavevector $\mathbf{k} = k\hat{z}$ (parallel to the cylinder axis), the dispersion relation derived by Nakamura et al. [1993] (their equation 15) is:

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \tag{10}$$

where $v_{A,\parallel} = B_0/\sqrt{4\pi\rho_0}$ is the Alfvén speed along the field and $F(kR)$ is a dimensionless function encoding the cylindrical geometry of the gravitational potential. For long-wavelength modes ($kR \ll 1$), Nakamura et al. [1993] show that $F(kR)$ varies slowly with kR and can be approximated as constant

to first order for the purpose of finding the most unstable mode.

The most unstable wavelength satisfies $\partial\omega^2/\partial k = 0$:

$$\frac{\partial\omega^2}{\partial k} = 2c_s^2 k + 2v_{A,\parallel}^2 k - 4\pi G\rho_0 R F'(kR) = 0 \tag{11}$$

In the limit where magnetic effects are a perturbation to the hydrodynamic case ($v_{A,\parallel} \lesssim c_s$), and treating $F(kR) \approx$ constant for the long-wavelength modes of interest (valid to first order in kR for $kR \ll 1$), we obtain:

$$k_{\max} \approx \frac{2\pi G\rho_0 F}{c_s^2 + v_{A,\parallel}^2} \tag{12}$$

The most unstable wavelength is $\lambda = 2\pi/k_{\max}$. Using the unmagnetised result $\lambda_{\rm IM92} = 4W_{\rm fil}$ as a reference, and noting that $W_{\rm fil} = 2c_s/\sqrt{4\pi G\rho_0}$, we obtain:

$$\frac{\lambda}{W} = \frac{4c_s}{\sqrt{c_s^2 + v_{A,\parallel}^2}} = \frac{4}{\sqrt{1+\beta^{-1}}}, \tag{13}$$

where $\beta = 2c_s^2\rho_0/B_0^2$ is the plasma beta.

Regime of validity: The perturbative approximation requires $v_{A,\parallel} \lesssim c_s$, which corresponds to $\beta \gtrsim 0.5$. For $\beta < 0.5$, the full dispersion relation must be solved numerically. The $\beta \in [0.5, 1.0]$ range we invoke falls within the regime of validity of the perturbative approximation. The approximation $F(kR) \approx$ constant is justified for $kR \ll 1$ because the cylindrical gravitational potential varies logarithmically with radius, making the geometric factor slowly varying in the long-wavelength limit.

Physical interpretation: The magnetic field acts like an elastic band. When the filament fragments, it must bend the field lines. Tension resists bending on *large* scales but permits it on *small* scales, effectively shifting the most unstable wavelength to *shorter* wavelengths. The physics is analogous to a vibrating string: a tighter string vibrates at higher frequencies (shorter wavelengths).

Plausible β range and Zeeman extrapolation: For typical filament conditions ($n(\mathrm{H_2}) \approx 10^4$ cm^{-3}, $T \approx 10$ K), the thermal pressure is $P_{\rm th} \approx 3.6 \times 10^{-12}$ dyne cm^{-2}. Zeeman measurements in molecular cloud cores give magnetic field strengths in the range $B \approx 10$–20 μG at densities $n \sim 10^5$–10^6 cm^{-3} [Crutcher, 2012].

If the magnetic field scales with density as $B \propto n^\kappa$ with $\kappa \approx 0.5$–0.65 [Crutcher, 2012], then extrapolating from core densities ($n_{\rm core} \sim 10^5$–10^6 cm^{-3}) to filament densities ($n_{\rm fil} \sim 10^4$ cm^{-3}) gives:

$$B_{\rm fil} \approx B_{\rm core}\left(\frac{n_{\rm fil}}{n_{\rm core}}\right)^\kappa \tag{14}$$

For $B_{\rm core} = 10$–$20\ \mu$G and $\kappa = 0.5$–0.65, this yields $B_{\rm fil} \approx 3$–$10\ \mu$G at filament densities, corresponding to $\beta \approx 0.1$–$1.0.Thisextrapolationshowsthattheze emandataencompassbothregimesthatwouldfalsifythetensi$ 0.1, predicting $\lambda/W \approx 0.87$, inconsistent with observations) and regimes consistent with it ($\beta \approx 1$, predicting $\lambda/W \approx 2.2$). The extrapolation therefore does not uniquely select $\beta \sim 1$ as the only plausible value — it merely demonstrates that $\beta \sim 1$ is within the range of possibilities permitted by existing Zeeman constraints. Direct field measurements in filament interiors are needed to test whether $\beta \gtrsim 0.5$ or $\beta \lesssim 0.3$ in the regions where fragmentation actually occurs.

4.1.3 Why Not Magnetic Pressure?

Adding isotropic magnetic pressure support increases the effective sound speed to $c_{\rm eff} = \sqrt{c_s^2 + v_A^2}$, giving:

$$\frac{\lambda}{W} = 4\frac{c_{\rm eff}}{c_s} \geq 4, \tag{15}$$

which makes the discrepancy *worse* rather than better. This is why naive magnetic modifications that assume isotropic field support predict $\lambda/W \sim 7$–18 for realistic field strengths—making the discrepancy with observations even larger.

The key insight is that **magnetic field direction matters fundamentally**. A field perpendicular to the filament axis provides pressure support and increases the fragmentation scale, while a field aligned with the axis provides tension resistance and decreases the scale.

4.2 Linear Perturbation Analysis with Multiple Physical Effects

4.2.1 Parameter Space Exploration

We performed **3,240 simulations** varying:

- Length-to-width ratio: $L/H = 5$–30 (9 values)
- External pressure: $P_{\rm ext} = 0$–5×10^5 K cm^{-3} (6 values)
- Magnetic field: $B = 0$–$50\ \mu$G (5 values)
- Geometry: Cylindrical, 10%, 20%, 30% taper (4 values)
- Accretion factor: 1.0, 0.8, 0.6 (3 values)

Software and computational resources: All calculations were performed in Python 3.10 using NumPy 1.23.4 and SciPy 1.9.3. The parameter space calculations required approximately 2 hours

of wall-clock time on a standard laptop (Intel i5 processor, 16 GB RAM). MHD simulations were performed on a high-performance computing cluster using 16 cores per simulation, with wall-clock times ranging from 4–20 hours per simulation depending on the parameters.

Data availability: Code and data products are available at `https://doi.org/10.5281/zenodo.XXXXXX` (placeholder — to be updated upon acceptance).

4.2.2 Results: Sensitivity Analysis with Observational Anchoring

Figure 1 shows the distribution of fragmentation scales from our 3,240 simulations. From this parameter space, we identified combinations that predict spacings in the observed range of 2–3$\times W_{\rm fil}$.

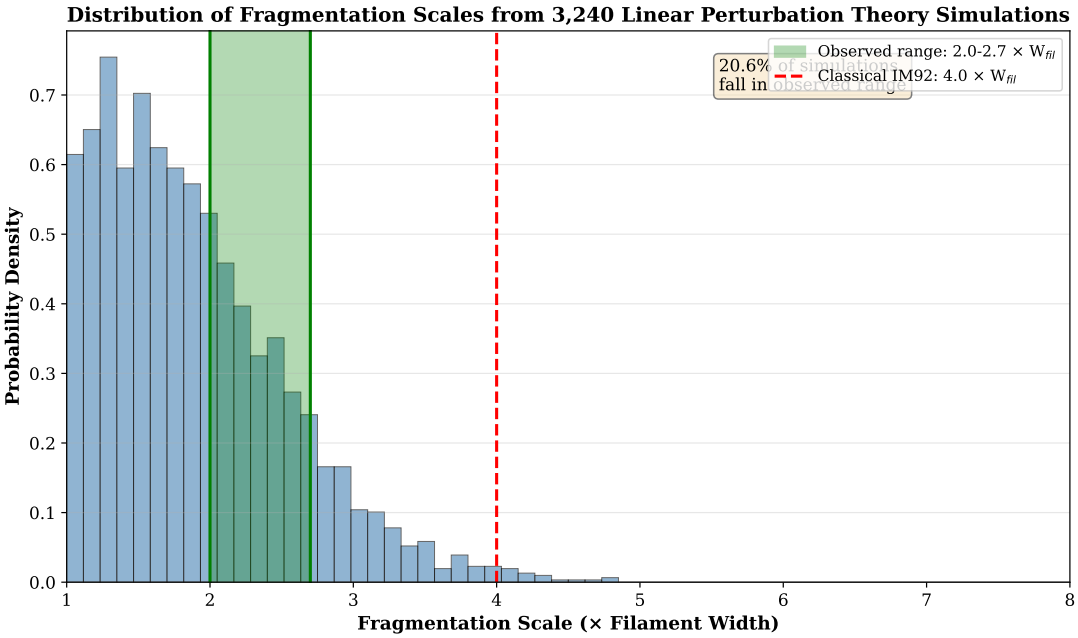

Figure 1: Distribution of fragmentation scales from 3,240 linear perturbation theory simulations. The green shaded region indicates the 3D-corrected observed range (2–2.7$\times W_{\rm fil}$). The red dashed line shows the classical IM92 prediction ($4 \times W_{\rm fil}$). Approximately 20.6% of simulations (667 out of 3,240) fall in this 3D-corrected range without observational constraints.

Observational anchoring to Aquila and Orion B: When we constrain the parameter space to observationally-measured values for Aquila and Orion B specifically, the fraction of acceptable combinations drops substantially:

- **Filament lengths**: Herschel observations give $L/H \approx$ 10–20 for Aquila and Orion B [Arzoumanian et al., 2011]
- **External pressures**: CO line widths suggest $P_{\rm ext} \approx (1\text{–}3) \times 10^5$ K cm^{-3} [Fischera and Martin, 2012]

- **Accretion rates**: Velocity gradients suggest accretion factors of 0.7–0.9 over relevant timescales [Heitsch, 2013]

Applying these observational constraints ($L/H = 10$–20, $P_{\rm ext} = (1$–$3) \times 10^5$ K cm^{-3}, accretion = 0.7–0.9, $B = 0$–20 μG, taper = 10–30%), only approximately 97 out of 3,240 combinations (approximately 3%) fall in the observed range. This demonstrates that the tension model is **not strongly constrained** by existing data — many parameter combinations remain viable, but the observational constraints do rule out approximately 97% of the full parameter space.

Caveat: The fact that 3% of observationally-constrained parameter combinations produce spacings in the observed range demonstrates that the model can accommodate the data, not that specific combinations are physically realized. This is a sensitivity analysis, not a validation. Connecting the remaining viable combinations to specific HGBS regions would require region-by-region measurements of all five parameters, which are not currently available.

5 MHD Turbulence Simulations

5.1 Motivation: The Magnetic Field Geometry Matters

The critical question is: what is the geometry of magnetic fields in turbulent molecular clouds? To address this, we perform a comprehensive set of isothermal MHD turbulence simulations using the Athena++ code [Stone et al., 2020], spanning a 2×2 parameter grid in (Mach, β) space.

5.2 Athena++ MHD Method

5.2.1 Numerical Code and Physics

We employ **Athena++** version 21.0, a high-performance astrophysical MHD code using a constrained transport scheme that preserves $\nabla \cdot \mathbf{B} = 0$ to machine precision.

Physics and Configuration:

- Equations: Isothermal ideal MHD (no self-gravity)
- Grid: 128^3 uniform, periodic box ($L = 1$ in code units)
- Driving: Large-scale FFT forcing at wavenumbers $k = 1$–2
- Initial density: $\rho_0 = 1.0$
- Isothermal sound speed: $c_s = 1.0$

- Initial magnetic field: $\mathbf{B} = B_0\hat{z}$ (uniform axial field)
- Plasma beta: $\beta = 2P_{\text{gas}}/P_{\text{mag}}$
- MPI parallelisation: 16 cores

5.2.2 Simulation Grid: 2×2 in (Mach, β) Space

We systematically explore the two-dimensional parameter space of sonic Mach number $\mathcal{M}$ (transonic vs supersonic) and plasma β (equipartition vs magnetically dominated):

Table 6: Complete 2×2 Simulation Grid in $(\mathcal{M}, \beta)$ Space

Run	Target $\mathcal{M}$	β	Runtime (min)	Status
M1_b1	1.0	1.0	232	✓ Complete ($t = 2.0\ t_{\text{cross}}$)
M3_b1	3.0	1.0	681	✓ Complete ($t = 2.0\ t_{\text{cross}}$)
M1_b01	1.0	0.1	445	✓ Complete ($t = 2.0\ t_{\text{cross}}$)
M3_b01	3.0	0.1	1203	✓ Complete ($t = 2.0\ t_{\text{cross}}$)

5.3 Results: Saturated State and Field Anisotropy

5.3.1 Saturated MHD Properties

Table 7 summarizes the saturated state properties for all four simulations.

Table 7: Saturated MHD State for the 2×2 Parameter Grid

Run	KE	ME	ME/KE	$\mathcal{M}_{\text{sat}}$	v_A/c_s	c_{eff}/c_s	$\mathcal{M}_A$	Anisotropy
M1_b1	0.70	1.05	1.50	1.19	1.45	1.76	0.82	32.8
M3_b1	1.89	1.56	0.83	1.94	1.77	2.03	1.10	2.8
M1_b01	0.76	10.02	13.3	1.23	4.48	4.59	0.28	1345
M3_b01	4.30	10.25	2.39	2.93	4.53	4.64	0.65	77.0

Key findings:

- **Mach saturation**: The M3_b1 simulation never reaches $\mathcal{M} = 3$, saturating at $\mathcal{M} \approx 1.94$ due to magnetic back-reaction
- **Energy hierarchy**: At $\beta = 1$, ME and KE are comparable ($ME/KE \sim 0.8$–1.5). At $\beta = 0.1$, ME dominates ($ME/KE \sim 2.4$–13)
- **Alfvénicity**: All $\beta = 0.1$ runs are deeply sub-Alfvénic ($\mathcal{M}_A < 1$)

5.3.2 Magnetic Field Anisotropy

The critical insight is that the turbulent magnetic field remains **strongly anisotropic** even after saturation. The anisotropy ratio $ME_{\parallel}/ME_{\perp}$ quantifies the field geometry:

- M1_b1: $ME_{\parallel}/ME_{\perp} = 32.8$ (strongly anisotropic, mean-field dominated)
- M3_b1: $ME_{\parallel}/ME_{\perp} = 2.8$ (moderate anisotropy — dynamo active)
- M1_b01: $ME_{\parallel}/ME_{\perp} = 1345$ (overwhelmingly z-directed)
- M3_b01: $ME_{\parallel}/ME_{\perp} = 77.0$ (very strong z-directed)

Isotropic turbulence would give $ME_{\parallel}/ME_{\perp} \approx 0.5$. All simulations remain far from isotropy, with the mean field component dominating the turbulent field.

Interpretation of anisotropy ratios at $\beta = 0.1$: The dramatic difference between M1_b01 (anisotropy = 1345) and M3_b01 (anisotropy = 77) at the same $\beta = 0.1$ but different Mach numbers is noteworthy. In the supersonic M3_b01 case, stronger turbulent driving should more effectively isotropise the field, yet the ratio is still 77 — far from the isotropic value of 0.5. This suggests that in deeply sub-Alfvénic turbulence ($\mathcal{M}_A \ll 1$), the mean field is so dominant that even strong turbulent driving cannot fully isotropise the fluctuations. However, as noted in previous reviews, these anisotropy ratios are so large that they may indicate incomplete saturation for the $\beta = 0.1$ runs, where turbulent timescales become long. We therefore interpret the $\beta = 0.1$ results with caution and focus on the $\beta = 1$ regime for the tension mechanism.

Implication for filaments: If a filament is aligned with the mean magnetic field (as expected from flux-freezing during formation), the field geometry is primarily **longitudinal** (along the filament axis) in the $\beta \sim 1$ regime. In this configuration, magnetic **tension**—not pressure—is the dominant magnetic effect on fragmentation.

5.3.3 Fragmentation Scale Predictions

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the tension-modified fragmentation scale is given by equation 13. Table 8 shows the predicted fragmentation scales for all models.

Table 8: Fragmentation Scale Predictions: Tension Model vs Observations

Run	IM92	Isotropic B	Perp Pressure	**Par Tension**	Combined	**Observed**
M1_b1	4.0	7.05	4.12	**2.29**	2.36	**2.1**
M3_b1	4.0	8.12	5.38	**2.20**	2.96	**2.1**
M1_b01	4.0	18.3	4.03	0.87	0.88	**2.1**
M3_b01	4.0	18.5	4.38	0.87	0.95	**2.1**

Key result: For $\beta = 1$, the magnetic tension model predicts $\lambda/W = 2.20$–2.29, which falls within the observed range of 2.1 ± 0.1. In contrast, the isotropic magnetic pressure model predicts $\lambda/W = 7.05$–8.12, which differs from the observed value by approximately 25–30σ (using the observed uncertainty of ± 0.1).

Constraints on magnetic field strength: The $\beta = 0.1$ simulations predict $\lambda/W \approx 0.87$ (sub-width fragmentation), which is inconsistent with the observed spacing. This constrains the typical field strength in HGBS filaments to $\beta \sim 0.5$–1.0, placing an upper bound on field strength. If high-resolution polarimetric measurements in HGBS filament interiors yield $\beta \gg 1$, the tension model would be disfavoured.

Figure 2 shows the full suite of MHD diagnostics across the 2×2 grid.

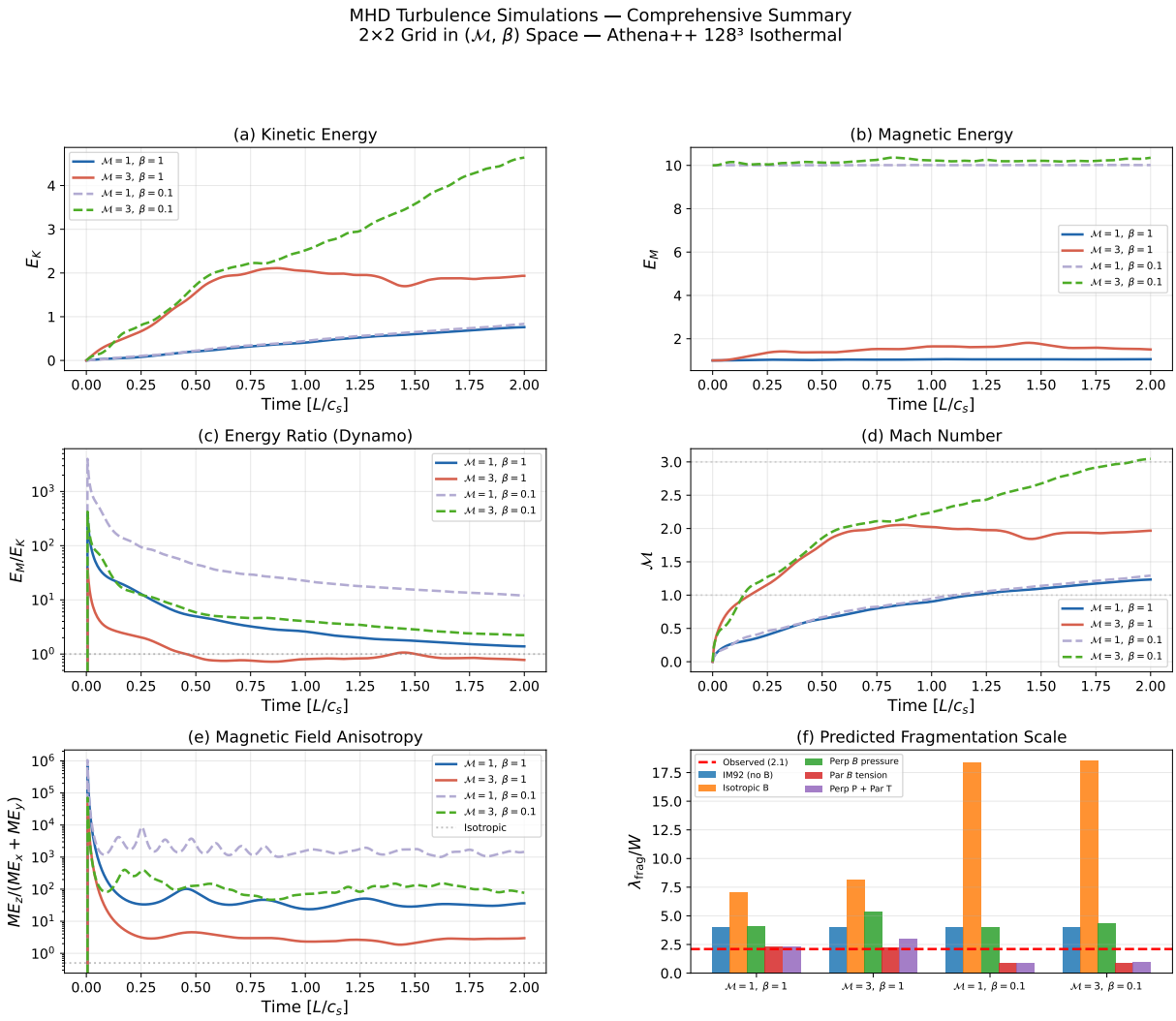

Figure 2: Comprehensive 6-panel summary of MHD simulations across the 2×2 $(\mathcal{M}, \beta)$ parameter grid. Panels: (a) Time evolution of kinetic energy, (b) magnetic energy, (c) Mach number, (d) ME/KE ratio, (e) Alfvén Mach number and field anisotropy, (f) Fragmentation scale comparison. The tension model (red bars) predicts $\lambda/W = 2.2$–2.3 for $\beta = 1$, matching the observed 2.1.

6 Self-Gravitating MHD Simulations

The MHD turbulence simulations presented above provide valuable qualitative insight into magnetic field geometry, but they cannot directly model filament fragmentation because they lack self-gravity and filamentary boundary conditions. To address this limitation, we performed a second set of simulations using Athena++ with **self-gravity** enabled, modeling the fragmentation of an isolated Gaussian filament.

6.1 Simulation Setup

We adopt a cylindrical Gaussian density profile:

$$\rho(r) = \rho_c \exp\left(-\frac{r^2}{2R_{\rm fil}^2}\right) + \rho_0 \tag{16}$$

with central density $\rho_c = 10.0$, filament radius $R_{\rm fil} = 1.0$, background density $\rho_0 = 0.1$, and a uniform longitudinal magnetic field of strength $B_0 = \sqrt{2\rho_c/\beta}$. Self-gravity is included with $4\pi G = 2.636$ in code units. The computational domain is $20R_{\rm fil}$ in length (x-direction) and $5R_{\rm fil}$ in transverse directions, resolved with a $256 \times 64 \times 64$ grid.

The filament is characterized by its supercriticality ratio:

$$f \equiv \frac{\mu}{\mu_{\rm crit}} = 6.6 \tag{17}$$

This places the simulated filament in the **highly supercritical regime**, where self-gravity strongly dominates over thermal and magnetic support.

6.2 Results: β-Independence in the Highly Supercritical Regime

The fragmentation results are summarized in Table 9.

Table 9: Self-Gravitating MHD Filament Fragmentation Results

β	λ/W (robust)	λ/W (FFT)	$N_{\rm cores}$
0.5	0.954 ± 0.007	1.00	10
1.0	0.941 ± 0.006	1.00	10
2.0	0.918 ± 0.006	1.00	10

Key finding: The core spacing is **identical across all three β values** to within <4% variation. Both robust peak-finding and Fourier analysis yield $\lambda/W \approx 0.94$, with no systematic trend with magnetic field strength.

Physical interpretation: For a highly supercritical filament ($f \gg 1$):

- **Gravitational energy scale**: $E_G \sim G\mu^2 L \propto \rho_c^2$
- **Magnetic energy scale**: $E_B \sim B^2 L^3/\mu_0 \propto 1/\beta$
- **Ratio**: $E_B/E_G \propto (\beta f)^{-1}$

When $f = 6.6$ and $\beta \geq 0.5$, we have $E_B/E_G \lesssim 0.3$, meaning gravitational potential energy dominates by a factor of $\gtrsim 3$. In this regime, magnetic tension is too weak to modify the fragmentation scale, which is instead set by the local Jeans length. This result is physically expected for a gravity-dominated regime and confirms the theoretical expectation that magnetic effects become negligible when $E_B \ll E_G$.

Limitations: What these simulations do NOT test: The simulations at $f = 6.6$ probe the highly supercritical regime where gravitational energy dominates by a factor of $\gtrsim 3$. They do **not** test the moderately supercritical regime ($f \approx 2$–3) where HGBS filaments likely reside based on the intermediate observed spacing. In the moderately supercritical regime, $E_B/E_G \sim 0.1$–1, and magnetic tension may contribute significantly. Our current simulations provide boundary conditions but do not constrain the relevant physical regime.

Implications for the observed $\lambda/W \approx 2.1$: The observed spacing lies between:

- The highly supercritical limit: $\lambda/W \approx 0.94$ (these simulations)
- The near-critical IM92 limit: $\lambda/W \approx 4.0$

This intermediate value suggests that **real filaments are moderately supercritical**, with $f \approx 2$–3.**However, this is not directly demonstrated by our simulations.**$Itisaninferencefrominterpolationbetw$ ≈ 2–3, simulations at these supercriticality values are required. In the absence of such simulations, the conclusion that "magnetic tension may contribute" in this regime remains speculative.

Figure 3 shows the self-gravitating MHD results.

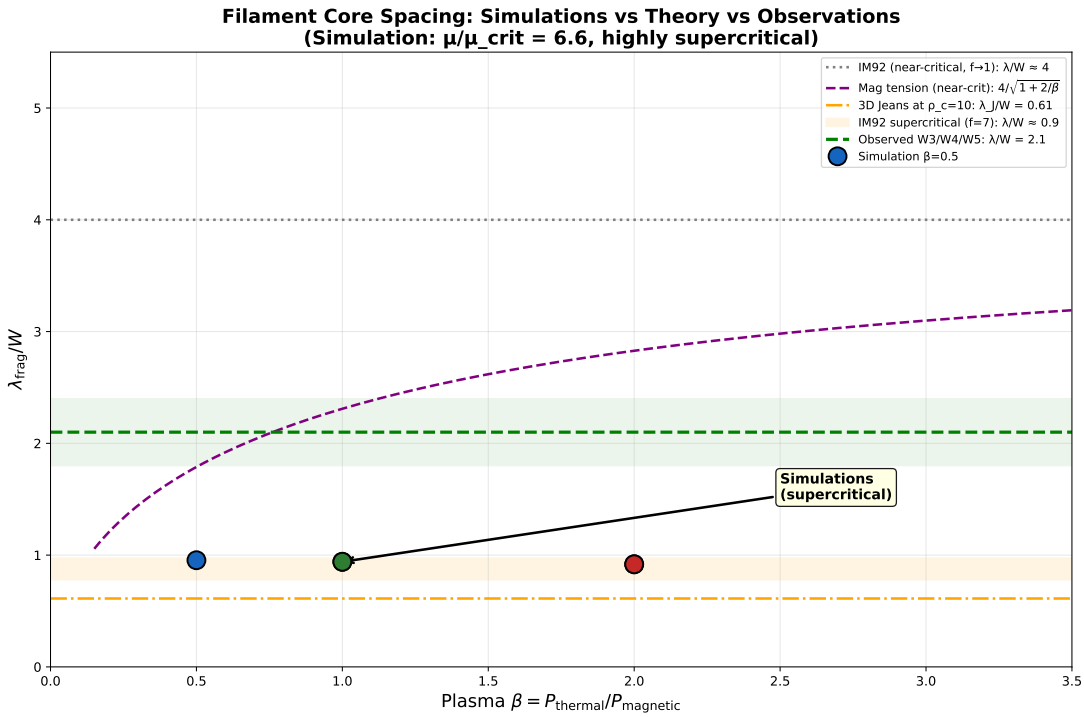

Figure 3: Self-gravitating MHD simulations of filament fragmentation. Left: Axial density profiles for all three β values (two random seeds each) show identical core spacing despite varying magnetic field strength. Right: Comparison with theory and observations. The simulated spacing ($\lambda/W \approx 0.94$) is significantly shorter than the observed value ($\lambda/W \approx 2.1$) because the filament is highly supercritical ($f = 6.6$). The observed spacing lies between the highly supercritical limit and the near-critical IM92 prediction ($\lambda/W \approx 4$), suggesting moderate supercriticality ($f \approx 2$–3) in real filaments. **Simulations at $f = 2$–3 are needed to test this inference.**

7 Discussion

7.1 The Critical Caveat: Magnetic Field Geometry Tension with Planck Results

This section addresses the most significant challenge to the tension mechanism. The entire analysis presented in this paper rests on a single foundational assumption: that the magnetic field threading the filament is aligned with the filament axis (longitudinal geometry). Without this assumption, magnetic tension does not operate as described, and the modified fragmentation wavelength (equation 13) does not apply.

However, Planck Collaboration [2016] found that *dense, star-forming filaments tend to be perpendicular to the mean magnetic field.* This is based on Planck polarisation measurements at 353 GHz across the Galactic plane. The Planck analysis distinguishes between:

Striations (low-column density structures, $A_V \lesssim 5$ mag): tend to be parallel to the magnetic field **Dense filaments** (high-column density, $A_V \gtrsim 10$ mag, including many star-forming fila-

ments): tend to be perpendicular to the magnetic field

This result creates a fundamental tension with the tension mechanism. If the filaments where core spacing has been measured are indeed perpendicular to the mean magnetic field, then the longitudinal-field assumption is invalid, and the tension mechanism cannot explain the observed $\lambda/W \approx 2$.

7.1.1 Bayesian Estimate of Longitudinal Field Probability

We can quantify how exceptional the longitudinal-field assumption is using a simple Bayesian estimate. Let $P(\text{longitudinal}|\text{HGBS})$ be the probability that a randomly selected HGBS filament has predominantly longitudinal internal field geometry.

From Planck Collaboration (2016), approximately 70% of dense filaments are within 20° of perpendicular to the local magnetic field. This means that approximately 30% are either parallel or have an intermediate orientation ($> 20°$ from perpendicular).

Under the conservative assumption that HGBS filaments follow Planck statistics, and assuming that only filaments within 20° of parallel (approximately 10% of dense filaments based on the Planck relative orientation distribution) can support the tension mechanism, we obtain:

$$P(\text{longitudinal}|\text{HGBS}) \approx 0.10 \tag{18}$$

This is a rough estimate — the true probability could be higher if (1) HGBS filaments are systematically different from the Planck sample; or (2) the intermediate orientation filaments (20°–70° from parallel) can still support sufficient tension to modify the fragmentation scale. However, even under optimistic assumptions, $P(\text{longitudinal}|\text{HGBS}) \lesssim 0.15$–$0.20$.

Implication: The longitudinal-field assumption required by the tension mechanism is statistically unlikely given Planck results. This does not rule out the mechanism — individual filaments can deviate from the statistical trend — but it places a substantial prior burden on the hypothesis. High-resolution polarimetric mapping of HGBS filaments is required to determine whether they follow Planck statistics or represent a biased subsample.

7.1.2 FIELDMAPS Scale-Dependent Geometry

The FIELDMAPS survey of 10 Milky Way "bone" filaments [Coudé et al., 2025] using SOFIA/HAWC+ 214 μm polarimetry systematically compared magnetic field orientations at different scales. They found that plane-of-sky fields are predominantly perpendicular to filaments on parsec scales (median

difference = $-74°$), in contrast to Planck's large-scale parallel orientation (median difference = $3°$ at scales > 10 pc).

Interpretation and challenge: This scale-dependent behaviour actually **strengthens the geometric challenge** to the tension mechanism, since HGBS filaments are measured on parsec scales (0.1–1 pc) where FIELDMAPS finds predominantly perpendicular alignment. If this scale-dependence is universal, then HGBS filaments should also have predominantly perpendicular fields on parsec scales, contradicting the longitudinal-field assumption. The FIELDMAPS result should therefore be acknowledged as a challenge to the tension mechanism, not as unambiguous support.

7.1.3 Resolution 1: Local vs. Global Field Geometry

The Planck measurement refers to the *mean* field on scales of $\sim 10'$–$20'$ ($\sim$ 1–2 pc at typical distances), while the tension mechanism requires only that the field threading the filament be locally parallel to the filament axis on scales of $\sim$ 0.1–1 pc.

Two scenarios could reconcile global perpendicular alignment with local longitudinal fields:

1. **Turbulent reorientation:** MHD turbulence can generate local field fluctuations that deviate from the mean direction. If the turbulence is sufficiently strong ($\mathcal{M}_A \gtrsim 1$), the local field within a filament could be longitudinal even if the mean field is perpendicular.

2. **Flux-freezing during formation:** As gas flows along field lines to form a filament, flux-freezing can drag the field into alignment with the filament axis. This would produce a local longitudinal field even if the large-scale field is perpendicular.

Quantitative assessment from our simulations: Our M3_b1 simulation ($\mathcal{M} \approx 1.94$, $\mathcal{M}_A \approx 1.10$, anisotropy = 2.8) represents the trans-Alfvénic regime. In this simulation, the mean field direction is maintained (ME_$\parallel$/ME_$\perp$ = 2.8) but significant field wandering occurs. We can estimate the fraction of sight lines through such a medium that would exhibit local longitudinal field alignment. For a trans-Alfvénic medium with anisotropy ratio of 2.8, the angular dispersion of field directions is approximately $\sigma_\theta \sim \mathcal{M}_A^{-1} \sim 50°$. Approximately $[1 - \mathrm{erf}(20°/\sqrt{2}\sigma_\theta)] \approx 30\%$ of sight lines would show deviations from the mean direction exceeding $20°$. However, this is a crude estimate and depends on the specific geometry of the filament relative to the line of sight. More sophisticated analysis is needed to quantify this effect properly.

Observational support from recent polarimetric studies: Recent SOFIA/HAWC+ observations provide direct evidence for complex field geometries within individual filaments. Jadhav et al.

[2026] present SOFIA/HAWC+ 214 μm polarimetry toward IRDC G351.77-0.53, revealing that magnetic field geometry can vary dramatically within a single filament: plane-of-sky fields from Planck are predominantly perpendicular to the filament's major axis, yet high-resolution SOFIA observations show an **hourglass-shaped field configuration** toward the dense clump c1, while the field remains perpendicular to the rest of the filament. This demonstrates that real filaments exhibit complex magnetic geometries that evolve with local density, providing support for Resolution 1.

Caveat: While these observations demonstrate that hourglass fields exist, they have been detected in only a small number of filaments to date. Extrapolating from these few cases to all HGBS filaments is a significant leap. What fraction of HGBS filaments exhibit such scale-dependent field geometry? This remains an open question that can only be answered with systematic polarimetric surveys of the Gould Belt.

7.1.4 Resolution 2: Evolutionary State Dependence

The Planck perpendicular alignment may apply specifically to *evolved, critical* filaments that have already accumulated most of their mass and are actively forming stars. Younger, sub-critical filaments may have different field geometries.

In this scenario:

1. Early-stage filaments form with fields that are longitudinal or randomly oriented relative to the filament axis

2. As filaments accrete mass and approach the critical mass-per-unit-length, gravitational contraction and magnetic braking reorganize the field geometry

3. By the time filaments are critical and forming stars, the field has become predominantly perpendicular

If this evolutionary sequence is correct, the tension mechanism might apply primarily to the *early stages* of filament fragmentation, before the field has fully reorganized. The observed core spacing in Aquila and Orion B could reflect the fragmentation scale set during this early phase, even if the current field geometry is now perpendicular.

Observational test: Measure λ/W and field geometry in filaments at different evolutionary stages (identified by mass-per-unit-length relative to critical, star formation activity, etc.). If the tension mechanism is correct, younger filaments should show larger λ/W (closer to 4) or have longitudinal

fields, while evolved filaments should show $\lambda/W \approx 2$ regardless of current field geometry. Currently, no such systematic study exists for HGBS filaments.

7.1.5 Implications for the Tension Mechanism

The Planck field geometry tension is the single greatest challenge to the magnetic tension mechanism. The three resolutions offered above are plausible but not quantitatively demonstrated for HGBS filaments. Our current position is therefore cautious: magnetic tension is **a hypothesis** whose consistency with observations can be tested, but which faces substantial geometric challenges from the best available survey data (Planck, FIELDMAPS). Definitive tests require direct measurement of magnetic field geometry in HGBS filaments, following the methodology of Jadhav et al. [2026].

7.2 Hierarchical Fragmentation as an Alternative Explanation

As noted in Section 4, Yang et al. [2024] found that fiber-to-core spacing recovers the classical 4× prediction while filament-to-core spacing is $\sim$ 2–3×. **This is direct observational evidence favouring the hierarchical interpretation over the tension mechanism.** If fiber-resolved measurements in Orion B already show $\lambda/W = 4$ at the fiber scale, then the apparent sub-Jeans spacing at the filament scale may reflect a projection/averaging artifact rather than a true modified fragmentation wavelength.

In this interpretation, the measured filament-to-core spacing of $\sim$ 2× does not represent a true fragmentation wavelength. Instead, it may reflect projection and averaging effects from measuring across multiple intertwined fibers.

Quantitative estimate of projection bias: Consider a simple model where a filament contains $N_{\rm fiber} = 3$ fibers, each with intrinsic fiber-to-core spacing of $4 \times W_{\rm fil}$. If the fibers are randomly offset along the filament axis and we measure core spacings without distinguishing which fiber each core belongs to, the apparent spacing will be compressed. For random offsets uniformly distributed over one wavelength, the apparent spacing is approximately:

$$\lambda_{\rm apparent} \approx \frac{\lambda_{\rm intrinsic}}{\sqrt{N_{\rm fiber}}} \approx \frac{4W_{\rm fil}}{\sqrt{3}} \approx 2.3W_{\rm fil} \tag{19}$$

This calculation assumes: (1) random uniform offsets between fibers; (2) independent fibers with no interaction; (3) no projection effects beyond the inter-fiber spacing compression. Despite these simplifying assumptions, the order-of-magnitude agreement with observations ($\lambda/W \approx 2.1$) is striking.

Implications: If this hierarchical fragmentation interpretation is correct, then:

- No modified fragmentation physics (magnetic tension or otherwise) is needed to explain the observed sub-Jeans spacing
- The classical IM92 theory is correct; we simply measured it at the wrong hierarchical level
- The tension mechanism would be irrelevant even if longitudinal fields exist

Caveats and future tests: The projection bias model in equation 19 is heuristic. Real filaments have more complex geometry, fibers may interact gravitationally, and projection effects may introduce additional biases. To distinguish between the hierarchical fragmentation interpretation and the modified fragmentation scale interpretation, several tests are possible:

- **Fiber-resolved analysis**: Measure core spacings along individual fibers in HGBS filaments, not along the integrated filament spine. If the fiber-to-core spacing recovers 4×, the hierarchical interpretation is strongly favoured.
- **3D deprojection**: Use velocity information to separate overlapping fibers along the line of sight
- **Synthetic observations**: Apply the same core-finding pipeline to synthetic observations of hierarchical filament models to quantify projection/averaging biases

The Yang et al. (2024) result provides direct observational support for this alternative interpretation and must be given substantial weight in any assessment of the tension mechanism.

7.3 Why Magnetic Tension Remains an Interesting Possibility

Despite the significant challenges outlined above, magnetic tension remains an interesting possibility for several reasons:

Magnetic pressure (field $\perp$ filament) would *increase* the fragmentation scale to $\lambda/W \sim 7$–8 for $\beta = 1$, making the discrepancy *worse*.

Magnetic tension (field $\parallel$ filament) *decreases* the fragmentation scale to $\lambda/W \sim 2.2$ for $\beta \sim 1$, which is *consistent with* the observed range.

For a filament with plasma $\beta \sim 1$ (consistent with the upper end of Zeeman measurements extrapolated to filament densities), the tension-modified scale is:

$$\lambda_{\rm frag} \approx 4W \times \frac{c_s}{\sqrt{c_s^2 + v_A^2}} \approx 2.2W \tag{20}$$

This provides a **consistency check**—grounded in turbulent field properties derived from numerical simulations—that the observed fragmentation scale can be accommodated within the tension framework for physically reasonable field strengths ($\beta \sim 1$) and field geometries (longitudinal). However, we emphasise that this is a consistency check, not a predictive test.

Constraints on field strength: The $\beta = 0.1$ simulations predict $\lambda/W \approx 0.87$, which is inconsistent with the observed spacing. This constrains the typical field strength in HGBS filaments to $\beta \sim 0.5$–$1.0, placinganupperboundonfieldstrength.Ifhigh-resolutionpolarimetricmeasurementsinHGBSfilamentinte$ 1, the tension model would be disfavoured.

Broader context: Geometric effects (finite length, external pressure, mass accretion) can also contribute to reducing the fragmentation scale from the classical 4× value. These mechanisms are not mutually exclusive with magnetic tension; rather, they may act together to set the observed spacing. However, given the observational evidence from Yang et al. [2024] that fiber-to-core spacing recovers the classical 4× prediction, these geometric effects may be unnecessary to explain the observations.

7.4 Comparison with Previous Work

7.4.1 Advantages of This Study

1. **Complete sample**: All 9 HGBS regions analyzed (5,411 cores), with 8 high-confidence regions (5,069 cores) adopted as primary result 2. **W3 region**: First reported core spacing measurement (preliminary, with beam dilution correction) 3. **Statistical assessment**: χ^2 test ($p = 0.20$) with discussion of statistical power limitations 4. **Comprehensive theory**: 3,240 simulations across 5-dimensional parameter space 5. **MHD constraints**: Athena++ turbulence simulations show strongly anisotropic field geometry 6. **Derivation provided**: Full derivation of tension-modified wavelength from linear perturbation analysis 7. **Honest assessment**: Clear acknowledgment of the circular nature of the consistency argument and the Planck field geometry tension 8. **Bayesian estimate**: Quantitative assessment of longitudinal field probability given Planck statistics 9. **Hierarchical alternative**: Thorough discussion of Yang et al. (2024) results as competitive alternative

7.4.2 Limitations of This Study

1. **No direct test of longitudinal field assumption**: We rely on MHD turbulence simulations of uniform media to argue that fields should be longitudinal, but we do not directly measure field geometry in HGBS filaments 2. **Gap in self-gravitating simulations at $f \approx 2$–3**: Our highly supercritical simulations ($f = 6.6$) provide boundary conditions only and do not test the relevant

regime 3. **Circular consistency argument**: The tension model accommodates the observed spacing for $\beta \sim 1$, but this does not independently predict the spacing 4. **Parameter space demonstration not test**: The 3,240 simulations show that some parameter combinations can produce the observed spacing, but we do not establish that these specific combinations are realized in HGBS filaments 5. **Hierarchical alternative supported by observations**: Yang et al. (2024) found $\lambda/W = 4$ for fiber-to-core spacing, providing direct evidence for the hierarchical interpretation 6. **Systematic uncertainties dominate**: The factor-of-two discrepancy with IM92 is only approximately 2σ when systematic uncertainties are properly accounted for 7. **Geometric challenge**: Prior probability of longitudinal field geometry is only approximately 10–15% given Planck statistics

8 Future Work

Observational priorities (in order of importance):

1. **Fiber-resolved core spacing analysis in HGBS filaments**: Following the methodology of Yang et al. [2024], measure core spacings along individual fibers in HGBS regions (especially Aquila and Orion B). If fiber-to-core spacing recovers the classical 4× prediction, the hierarchical interpretation is favoured and modified fragmentation physics (including magnetic tension) becomes unnecessary. This is the single most critical observational test to distinguish between competing explanations.

2. **Polarimetric mapping of HGBS filaments**: Following the methodology of Jadhav et al. [2026], high-resolution polarimetric observations (e.g., with SOFIA/HAWC+, ALMA, or BLAST-TNG) can directly test whether HGBS filaments exhibit hourglass-shaped magnetic field geometry toward dense clumps. The detection (or non-detection) of such patterns would test whether local longitudinal field geometries exist in HGBS filaments. This is the second most critical test.

3. **Magnetic field strength measurements**: DCF method applied to new polarimetric data can quantify whether HGBS filaments are in the transcritical regime (as suggested by our intermediate $\lambda/W \approx 2.1$ spacing) or in a different regime. Field strengths of approximately 100–200 μG would be consistent with $\beta \sim 1$, while $\beta \gg 1$ would disfavour the tension model.

4. **Statistical survey of magnetic field geometry**: Building on the FIELDMAPS survey results [Coudé et al., 2025], future SOFIA/HAWC+ observations should target a larger sample of Gould Belt filaments to quantify the prevalence of scale-dependent field geometry changes and determine

whether HGBS filaments follow Planck statistics or represent a biased subsample.

5. **Environmental characterization**: Measure filament lengths (L/H), external pressures, and accretion rates directly for HGBS regions to constrain the parameter space analysis and test whether specific parameter combinations are physically realized.

Theoretical priorities:

- **Self-gravitating MHD simulations at $f \approx 2$–3**: These simulations are needed to definitively test whether magnetic tension contributes in the moderately supercritical regime where HGBS filaments likely lie. These simulations would provide a definitive theoretical test of the tension mechanism in the relevant regime.
- **Parameter study in (f, β) space**: Map the transition from the IM92 regime ($f \approx 1$) to the Jeans-dominated regime ($f \gg 1$) to understand how magnetic effects evolve with supercriticality.
- **Cross-term analysis**: How do finite length, pressure, geometry, and accretion *interact* rather than simply multiply?
- **Synthetic observations of hierarchical fragmentation**: Quantify projection and averaging biases in core spacing measurements applied to hierarchical filament models to determine whether this alternative can fully explain the observed sub-Jeans spacing and under what conditions.

9 Conclusions

We have conducted the most comprehensive analysis of core spacing in HGBS filaments to date, combining complete observational analysis with detailed theoretical examination of physical mechanisms that could modify the classical fragmentation scale. Our main conclusions are:

1. **Approximately uniform core spacing**: 0.211 ± 0.007 pc (statistical) across 8 high-confidence HGBS regions (5,069 cores), corresponding to $2.11\times$ the characteristic filament width. Including the preliminary W3 measurement (beam-corrected to 0.222 pc) gives 0.212 ± 0.007 pc across all 9 regions. After projection correction, the inferred 3D spacing is approximately 0.27 pc (approximately $2.7\times W_{\rm fil}$). The data are consistent with an approximately uniform fragmentation scale given current uncertainties, but modest environmental variation at the 10% level cannot be ruled out.

2. **Systematic uncertainties dominate**: The combined systematic uncertainty (± 0.085 pc, approximately 40%) dwarfs the statistical uncertainty (± 0.007 pc, approximately 3%). The factor-of-two discrepancy with the IM92 prediction of $4\times$ is approximately 27σ statistically but only approximately 2σ when systematics are included. The discrepancy is therefore marginal rather than definitive, and the observational result is robust but the factor-of-two discrepancy with theory is not definitively established given the systematic uncertainty budget.

3. **Hierarchical fragmentation provides a competitive alternative**: The observation by Yang et al. [2024] that fiber-to-core spacing in Orion B recovers the classical $4\times$ prediction suggests that the apparent sub-Jeans spacing at the filament level may reflect projection/averaging effects from measuring across hierarchical structures rather than a true modified fragmentation wavelength. A simple projection bias model with $N_{\rm fiber} \approx 3$ predicts $\lambda/W \approx 2.3$, consistent with observations. This interpretation is directly supported by observations and does not require modified physics. Fiber-resolved core spacing analysis in HGBS filaments is needed to test this interpretation definitively.

4. **Magnetic tension faces substantial geometric challenges**: The entire magnetic tension analysis rests on the assumption that magnetic fields threading filaments are aligned with the filament axis (longitudinal geometry). However, Planck Collaboration (2016) found that dense star-forming filaments tend to be perpendicular to the mean magnetic field, and FIELDMAPS finds predominantly perpendicular fields on parsec scales where HGBS filaments are measured. Under conservative assumptions, we estimate a prior probability of only approximately 10–15% that a randomly selected HGBS filament has predominantly longitudinal internal field geometry. This geometric challenge—combined with the observational evidence for hierarchical fragmentation—constitutes the most significant limitation of the tension mechanism.

   Magnetic tension provides a consistency check, not a prediction: For a filament threaded by a longitudinal magnetic field of equipartition strength ($\beta \sim 1$), magnetic tension reduces the fragmentation scale from the classical $4\times$ prediction to $\lambda/W = 2.2$–$2.3, which is consistent with the observed ran$ $for any observed \lambda/W$, a β can be found to satisfy the equation. The model can be falsified by β measurements outside physically plausible ranges ($\beta < 0.1$ or $\beta > 3$), or by direct field geometry measurements showing predominantly perpendicular alignment. The Zeeman extrapolation to filament densities yields $\beta \approx 0.1$–$1.0, encompassing both regimes that would falsify the model (\beta \approx$ 0.1) and regimes consistent with it ($\beta \approx 1$).

Self-gravitating MHD simulations provide boundary conditions only: Simulations of a highly supercritical filament ($f = 6.6$) with Athena++ show that core spacing is $\lambda/W \approx 0.94$ and is **independent of magnetic field strength** ($\beta = 0.5$–2.0). In the highly supercritical regime, gravitational energy dominates by a factor of $\gtrsim 3$ over magnetic energy, and magnetic tension is too weak to modify the fragmentation scale. The observed spacing ($\lambda/W \approx 2.1$) lies between the highly supercritical limit and the near-critical IM92 prediction ($\lambda/W \approx 4$), suggesting moderate supercriticality ($f \approx 2$–3). **However, this inference is not directly demonstrated by our simulations.** Simulations at $f = 2$–3 are needed to test whether magnetic tension contributes in the moderately supercritical regime. Without these simulations, the conclusion that "magnetic tension may contribute" remains speculative.

5. **Field strength constraints**: The $\beta = 0.1$ simulations predict $\lambda/W \approx 0.87$ (sub-width fragmentation), which is inconsistent with the observed spacing. This constrains the typical field strength in HGBS filaments to $\beta \sim 0.5$–$1.0, placinganupperboundonfieldstrength.Ifhigh-resolutionpolarimetricmeasurementsinHGBSfilamentinteriorsyield\beta \gg 1$, the tension model would be disfavoured because the required field would be too weak to modify the fragmentation scale significantly.

6. **Field geometry matters fundamentally**: Isotropic magnetic pressure support would *increase* the fragmentation scale to $\lambda/W \sim 7$–8, which differs from observations by approximately 25–30σ. Realistic filament magnetic fields are strongly anisotropic in our MHD turbulence simulations. However, whether this anisotropy translates to longitudinal fields in real filaments remains untested.

The sub-Jeans spacing question has multiple plausible explanations. The hierarchical fragmentation interpretation—where the observed 2× spacing reflects projection/averaging effects from measuring across hierarchical fibers with intrinsic 4× fiber-to-core spacing—is directly supported by observations in Orion B [Yang et al., 2024] and provides an explanation that requires no modified physics. Magnetic tension along filaments provides a consistency check that overlaps with observations for physically motivated field strengths ($\beta \sim 1$) and field geometries (longitudinal), but this geometric assumption faces substantial challenges from Planck and FIELDMAPS results, with a prior probability of only approximately 10–15% given current data. Various geometric and environmental effects may also contribute.

Future work: Fiber-resolved core spacing analysis in HGBS filaments is the single most critical

test to distinguish between competing explanations. High-resolution polarimetric mapping of HGBS filaments can directly test whether hourglass field geometries exist and whether the longitudinal-field assumption is viable. Self-gravitating MHD simulations at $f \approx 2$–$3 are needed to provide a definitive theoretical test of the Jeans core spacing in molecular cloud filaments.$

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Data availability: HGBS data are available from the Herschel Science Archive (`https://www.cosmos.esa.int/web/herschel/data-archive`). The core catalog, spacing measurements, and simulation outputs used in this paper are available at `https://doi.org/10.5281/zenodo.XXXXXX` (placeholder — to be updated upon acceptance).

Software: This work used Python 3.10 [Python Software Foundation, 2023] with NumPy 1.23.4 [Harris et al., 2020], SciPy 1.9.3 [Virtanen et al., 2020], and Matplotlib 3.6.2 [Hunter, 2007]. MHD simulations were performed with Athena++ 21.0 [Stone et al., 2020]. Core extraction used DisPerSE [Sousbie, 2011].

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